

ATMOSPHERIC DISPERSION SURROGATE RESEARCH

AERMOD emulator prototype - public technical note

Version 0.1 | 17 September 2026

Status: Independent research prototype. Not a regulatory replacement for AERMOD. The current evidence applies only to the restricted domain described in this note.

This note is a public summary of a larger private research record. It describes the question, the main scientific turns, the frozen validation results and the current limits. Large teacher datasets, model weights, controller code and the exact production residual library are not part of the public release.

The project uses AERMOD as the teacher/reference model. The objective is to emulate the AERMOD response inside a validated domain, not to claim that the surrogate predicts the real atmosphere better than AERMOD.

Independent project. Not affiliated with or endorsed by the U.S. Environmental Protection Agency.

1. Research question

AERMOD can generate detailed hourly concentration fields from source, meteorological and receptor information. The research question here is narrower: can a separate surrogate learn that response well enough to make fast field predictions for cases it did not see during development?

The target is not an annual maximum or a small set of summary statistics. The core target is the hourly receptor concentration field. Maxima, averages, contours and exceedance checks are treated as downstream quantities.

Simple view: source and meteorological state + plume-relative receptor geometry -> frozen surrogate -> hourly concentration field.

AERMOD remains the teacher, verification reference and regulatory model. The surrogate is useful only where the evidence shows that it is inside its validated domain.

2. Prototype boundary

The first prototype was deliberately kept small so that the source of error could be diagnosed instead of hidden inside many interacting model options.

- One fixed elevated point source.
- SO₂ treated as a passive pollutant for the prototype.
- Flat/rural terrain.
- No building downwash.
- No chemistry or deposition.
- Controlled receptor geometry expressed in plume-relative coordinates.
- Hourly concentration field as the main output.

This means the current result should not be read as a general AERMOD surrogate. It is evidence that the mapping can be learned inside one restricted physical and climatological domain.

3. How the teacher data are used

AERMOD is treated as a physics-response generator. Each teacher case links an atmospheric state and receptor geometry to an AERMOD concentration response. In the original full-year teacher set, the project generated more than 6.6 million receptor-hour rows.

Concentration was normalised by emission rate, C/Q , where emission-rate linearity was valid for this controlled case. A log transform was used during learning to reduce the effect of the very large dynamic range between near-zero and plume-centre concentrations while keeping the transformation reversible.

Train/test separation is done by meteorological time blocks or entire years. Random receptor-row splits are avoided because rows from the same hour share the same atmospheric state and would make the test unrealistically easy.

4. Why the spatial representation changed

The first important failure was not caused by the learner. It was caused by the way the AERMOD field had been sampled.

An early 10-degree angular receptor grid was too coarse for some narrow stable plumes. In those cases, the teacher grid itself did not contain enough spatial information to reconstruct the narrow peak reliably. Adding more trees to the machine-learning model could not solve that problem.

The project therefore moved to plume-relative Cartesian geometry using downwind and crosswind distance. Interpolation experiments were run before further model fitting so that representation error could be separated from machine-learning error.

Crosswind reconstruction worked well first. Longitudinal coverage and then two-dimensional near-source coupling each failed in separate tests. The response was targeted densification of the spatial support rather than an automatic increase in model complexity.

A fresh 2024 evaluation later exposed another crosswind-resolution weakness in narrow stable plumes. That year was then treated as spent development evidence rather than reused to claim that the repaired representation had passed an independent test.

5. Current surrogate structure

The public release keeps the exact production recipe private, but the current frozen prototype can be described at a useful level.

- A gradient-boosted field model using meteorological and plume-relative features.
- A learned longitudinal amplitude correction that addresses dilution/amplitude error seen in fresh-year testing.
- A strongly bounded local residual correction estimated from development data only.
- A separate support-distance calculation used to describe whether a prediction resembles the development support.

Several more complicated successor ideas were tested and rejected. Regime-specific residual experts, low-dimensional residual predictors and direct lateral corrections did not show enough robust improvement to replace the frozen prototype.

6. Validation design

The main validation rule is simple: once a year has been opened and used to change the model, it cannot be called fresh validation again.

Year	Role when first opened	What happened after opening
2024	Fresh representation test	Used as development evidence after a crosswind-resolution failure was diagnosed.
2025	Fresh error-mechanism test	Used as development evidence after amplitude/dilution error was diagnosed.
2022	Untouched architecture exam	Supported the architecture direction, then became spent development evidence.
2019	Untouched temporal confirmation	One-shot evaluation of the frozen model.
2018	Untouched temporal confirmation	Second one-shot evaluation of the same frozen model.

For the 2019 and 2018 tests, the student-side prediction package was fixed before the AERMOD concentration truth was opened. No validation-year retraining or post-hoc applicability-threshold tuning was allowed in the reported result.

7. Frozen 2018 and 2019 results

The same frozen surrogate was evaluated on 2019 and then 2018. The results were close in magnitude, which is useful because the two years were different hourly meteorological realisations.

Metric	2018	2019
Usable hours	8,620	8,531
Baseline whole-field normalised L1	0.1690417	0.1721767
Frozen surrogate whole-field normalised L1	0.1272474	0.1312534
Relative whole-field improvement	24.72%	23.77%
Baseline amplitude normalised L1	0.1221596	0.1228541
Frozen surrogate amplitude normalised L1	0.0767853	0.0808092
Relative amplitude improvement	37.14%	34.22%

The result supports repeatable same-domain temporal improvement. It does not mean that the surrogate is 24% “accurate,” and it is not a comparison against measured ambient concentration. The percentage is the relative reduction in one project error metric against a frozen internal baseline surrogate.

Extreme peaks

The strongest caution comes from the global maxima. In 2018, the AERMOD maximum was about 3.02 in the project C/Q scale while the surrogate produced about 7.11, at a different case/receptor. In 2019, the AERMOD maximum was about 3.10 while the surrogate produced about 6.61, again at a different location.

Interpretation: the surrogate improves the field as a whole, but it is not yet reliable for the extreme regulatory tail or for identifying the controlling receptor.

8. Are 2018 and 2019 really different?

Because the two validation years came from the same site, it was important to check that the second result was not simply a replay of nearly identical weather.

Comparison	Result
Same-calendar wind-speed correlation	0.043
Same-calendar friction-velocity correlation	0.145
Same-calendar mechanical mixing-height correlation	0.123
Median absolute wind-direction difference	75 degrees
Identical full dynamic states at comparable calendar hours	0
Exact rounded dynamic-state overlap across the years	about 0.15%

The years still share the same broad local climatology, so seasonal temperature and day/night behaviour are expected to be related. The correct claim is therefore same-domain temporal generalisation across materially different hourly meteorology, not transfer to a radically different climate or site.

9. What failed and what was learned

Several negative results changed the design.

- More trees gave diminishing returns, so the work moved from capacity to representation and residual structure.
- The 10-degree teacher grid missed narrow stable plumes, so the geometry moved to denser plume-relative Cartesian support.
- Longitudinal interpolation initially failed, which led to targeted x-direction densification.
- Two-dimensional composition failed near the source, showing that good one-dimensional components did not guarantee a good coupled field.
- Fresh 2024 stable cases failed, so crosswind support was refined and 2024 became spent evidence.
- Fresh 2025 error was dominated by amplitude, which led to residual amplitude correction.
- Several successor residual models failed frozen gates, so the simpler Phase2H prototype was kept.

This is why the current model is treated as a research milestone rather than a finished engine. The main value is not one model score; it is the sequence of tests that narrowed down what information and architecture actually mattered.

10. Current limitations

- The model has not reached extremely small field error across all receptors and hours.
- Extreme peak magnitude and peak location remain unreliable.
- The applicability/support score is informative but not an error guarantee.
- The validation evidence is from one broad site climatology.

- Complex terrain, downwash, multiple sources, other source types and chemistry/deposition remain outside the current validated scope.
- The surrogate is not an approved regulatory replacement for AERMOD.

These limits define the actual scientific boundary of the current release.

11. Next research boundary

The first narrow question has been answered well enough to stop small repeated tweaks on the same setup. The next useful experiment should expand one boundary at a time.

- Peak/tail modelling: understand why the controlling extremes remain unstable without tuning to known validation peaks.
- New site or climate: test whether the learned response transfers outside the current local climatology.
- Variable source physics: introduce a bounded range of release height, diameter, exit velocity and temperature.
- Only after those are stable: terrain, downwash, multiple sources and wider source classes.

12. Reproducibility and public/private split

The private research lab keeps the detailed evidence chain: teacher outputs, run manifests, frozen hashes, validation seals, failure records, model artifacts and operational controller history. The public repository deliberately contains a smaller set of material.

The public release includes the research question, methodology, validation tables, limitations, research history, a small synthetic input-schema example and this technical note. It does not include the trained model weights, full teacher library, exact residual library or production orchestration.

That split allows the work to be inspected and discussed without publishing every implementation detail before the project has made a longer-term IP and commercial decision.

References

U.S. Environmental Protection Agency. Support Center for Regulatory Atmospheric Modeling (SCRAM), Preferred and Recommended Models. AERMOD Modeling System code and documentation. Current AERMOD release information accessed September 2026.

<https://www.epa.gov/scram/air-quality-dispersion-modeling-preferred-and-recommended-models>

Internal source basis: *Generic AERMOD AI Engine - Complete Project History, Scientific Rationale, Experimental Record, Failures, Governance, and Validation*. Controlled reconstruction dated 17 September 2026. Detailed source register and run artifacts are retained in the private project archive.